

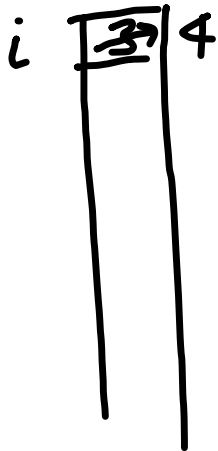
Targets/Goals:

Students will learn about and use math in programs.

Students will learn about floating point versus integer math (modulus).

Students will learn about random numbers in programs.

Math vs. Programming

 $i = i + 1$ math (never true) $i = i + 1 ;$ // programming $i = i + 1$
↑
"is assigned the value"

Math operators

int float	{	+	add
		-	subtract
		*	multiply
		/	divide
only int (long)	{	%	modulus

$$e = mc^2$$

$$e = m * c * c;$$

order of operations
is in effect

PEMDAS
↑
modulus

^ is not
exponent
↓
pow ()

```
static void Main (string[] args)
{
    int a = 3, b = 2, c = 7;
    double x = 3.1, y = 4.2, z = 6.8;
    :
    a = b + c; // a = 9
    Console.WriteLine(a);
    a = b * c; // a = 14
    a = c / b; // a =
    :
}
```

a

3

$c=7, b=2$
 $a = c / b;$ // ~~$a = 3.5$~~

2 | 7
 6
 —
 1

truncate (get rid of decimal)
 double $x = 2.0, y = 7.0;$

$a = y / x;$

← 3.5
 $a \rightarrow 3$

3.5

$b=2, c=7$

modulus
↓

finds
remainder
of
int division

$$a = c / b; \quad // a = 3$$

$$a = c \% b; \quad // \text{modulus}$$

3

2 | 7

6

R 1

Logical operators

	NOT
&	AND
	OR

(branching)

Compound operators

increment/decrement



+=

(add 1) ++ (c++)

-=

(sub 1) --

*=

/=

int = [% =

a += 7;

a = a + 7;

a = a + 1;

a += 1,
a++;

++a;

Input and output

```
static void Main(string[] args)
{
    Console.WriteLine("Hello, World!");

    const char sqrtSym = '\u221A'; // unicode square root symbol
    Console.WriteLine(sqrtSym + " " + 7);

    Console.Write("What is your name? >> ");
    String name = Console.ReadLine(); ← input string
    Console.WriteLine("Hello " + name);

    Console.Write("What is your age? >> ");
    int age = Convert.ToInt32(Console.ReadLine()); ← input int
    Console.WriteLine("age: " + age);

    Console.SetCursorPosition(10, 15);
    Console.Write(name);
    Console.SetCursorPosition(10, 16);
    Console.Write(age);
    Console.ReadKey(); // only need for external console
}
```

Math

```

static void Main(string[] args)
{
    Console.WriteLine("Hello, World!");
    int a = 7;
    int b = 3;
    double c = 7.8;
    double d = 3.4;
    int iAns = a + b;
    double fAns = c + d;

    Console.WriteLine($"a = {a}\t\tb = {b}\nc = {c}\t\td = {d}
\n");
    Console.WriteLine("a + b = " + iAns);
    Console.WriteLine($" {c} + {d} = {fAns}");

    /* Do subtraction, multiplication, division */

    iAns = a / b;
    fAns = c / d;
    Console.WriteLine("a / b = " + iAns);
    iAns = a % b;
    Console.WriteLine("a % b = " + iAns);
    Console.WriteLine($" {c} / {d} = {fAns}");

    iAns = a / (int) c; // typecasting (casting)
    fAns = b / d;
    Console.WriteLine("a / c = " + iAns);
    Console.WriteLine($" {b} / {d} = {fAns}");

    Console.ReadKey();
}

```

Random Numbers

```
class Program
{
    static Random rng;
    static void Main(string[] args)
    {
        Console.WriteLine("Hello, World!");
        rng = new Random();

        int dice = rng.Next(6);
        dice = 1 + dice;
        Console.WriteLine(dice);
        dice = rng.Next(1, 7);
        Console.WriteLine(dice);

        Console.ReadKey();
    }
}
```

By the end (exit):

Start a program using math and/or random numbers and push to GitHub.